# Machine Learning-Based Employee Attrition Prediction and Risk Scoring System

Palo Alto Networks — HR Analytics Platform

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| **Status** | Completed |
| **Dataset** | 1,470 Employee Records · 35 Features |
| **Models** | Logistic Regression · Random Forest · XGBoost |
| **Best ROC-AUC** | 81.0% (Logistic Regression) |

# Abstract

Employee attrition represents a significant operational and financial challenge for technology organizations. This paper presents the development and evaluation of a machine learning-based attrition prediction and risk scoring system designed for Palo Alto Networks. The system applies a multi-model classification pipeline — comprising Logistic Regression, Random Forest, and XGBoost — to a 1,470-record employee dataset, producing employee-level attrition probability estimates and interpretable risk categories. The best-performing model achieves a ROC-AUC of **0.810** with a recall of **74.5%** on attrition cases, demonstrating meaningful predictive capability beyond chance. Results are surfaced through an interactive Streamlit dashboard enabling HR stakeholders to explore individual employee risk profiles, department-level patterns, and model explanations using SHAP values and feature importance. The system is designed to support proactive, data-driven retention planning rather than reactive workforce management.


# 1. Introduction

## 1.1 Background

Employee attrition — the voluntary or involuntary departure of employees from an organization — imposes substantial direct and indirect costs. Tangible costs include recruitment, onboarding, and training; intangible costs include lost institutional knowledge, reduced team cohesion, and productivity disruption. For technology firms like Palo Alto Networks, where specialized talent is critical and competitive labor markets intensify turnover risk, the ability to anticipate and respond to attrition risk proactively is a strategic priority.

Traditional HR analytics has largely been retrospective: reporting on attrition rates after the fact, or identifying patterns in historical exit data. This approach provides limited time for intervention. A predictive system that identifies employees with elevated departure risk in advance enables HR teams to act before attrition occurs — through retention conversations, compensation adjustments, career development support, or targeted engagement programs.

## 1.2 Problem Statement

Palo Alto Networks faces three interconnected attrition challenges:

- **Unanticipated departures** — Resignations that arrive without warning, leaving insufficient time for knowledge transfer or counter-offers.
- **Loss of critical talent** — High-performing or specialized employees whose departure is disproportionately disruptive.
- **Reactive retention** — Counter-offer strategies deployed after an employee has already decided to leave.

## 1.3 Objectives

- Develop a machine learning pipeline that classifies employees as likely to attrite or remain.
- Generate calibrated attrition probability estimates for each employee.

- Assign employees to interpretable risk categories: Low, Medium, and High.
- Identify the features most strongly associated with predicted attrition.
- Provide an interactive dashboard enabling HR stakeholders to explore findings.

## 2. Dataset

### 2.1 Overview

The dataset contains employee-level records for Palo Alto Networks personnel — **1,470 records** and **35 attributes** spanning demographic, compensation, satisfaction, career, workload, and organizational dimensions. The target variable *Attrition* is binary (1 = left, 0 = stayed). The dataset exhibits class imbalance: approximately **16.1%** of employees attrited (237 records) versus 83.9% who remained (1,233 records).

### 2.2 Selected Key Features

| Feature | Description |
|---|---|
| MonthlyIncome | Monthly salary — top attrition predictor |
| OverTime | Overtime indicator — workload proxy |
| JobSatisfaction | Self-rated job satisfaction (1–4) |
| YearsSinceLastPromotion | Career advancement recency |
| YearsWithCurrManager | Manager relationship duration |
| NumCompaniesWorked | Prior employer count — mobility history |
| EnvironmentSatisfaction | Workplace environment rating (1–4) |
| WorkLifeBalance | Work-life balance self-rating (1–4) |

## 3. Methodology

### 3.1 Data Preprocessing

The preprocessing pipeline applied the following steps: **(1) Data Cleaning** — removed constant columns (EmployeeCount, EmployeeNumber, Over18, StandardHours) and converted categorical targets to binary integers. **(2) Label Encoding** — all remaining string columns encoded via LabelEncoder, appropriate for the tree-based models used. **(3) Standard Scaling** — all numerical features standardized to zero mean and unit variance using StandardScaler, applied independently to train and test sets. **(4) Stratified Train-Test Split** — 80% training (1,176 records) / 20% test (294 records) with class-ratio preservation. **(5) SMOTE** — Synthetic Minority Oversampling Technique applied after splitting to the training set only, balancing classes to 988 records each (post-SMOTE training size: 1,976 records).

### 3.2 Feature Engineering

Four composite features were constructed beyond the raw variables:

| Feature | Formula / Logic | Rationale |
|---|---|---|
| **IncomeExperienceRatio** | MonthlyIncome / (TotalWorkingYears + 1) | Compensation relative to experience — underpaid tenure signals risk |
| **PromotionDelayFlag** | 1 if YearsSinceLastPromotion ≥ 4 | Flags prolonged career stagnation |
| **EngagementScore** | Mean of 5 satisfaction dimensions | JobSatisfaction, EnvSatisfaction, RelSatisfaction, WLB, JobInvolvem |
| **WorkloadStressFlag** | 1 if OverTime = Yes | Explicit workload pressure indicator derived from OverTime |

## 3.3 Model Development

**Logistic Regression**: Linear classifier providing an interpretable baseline. Trained with class_weight='balanced' and max_iter=1000. Coefficients provide directional insight into feature-attrition relationships.

**Random Forest**: Bagged ensemble of 300 decision trees. Trained with class_weight='balanced', min_samples_leaf=2. Captures non-linear relationships and feature interactions. Provides Gini-based feature importance.

**XGBoost**: Gradient-boosted ensemble with n_estimators=400, learning_rate=0.05, max_depth=6. scale_pos_weight set to inverse class ratio for imbalance handling.

## 3.4 Risk Scoring Framework

Attrition probabilities are mapped to three risk categories using configurable thresholds. Default thresholds are **30%** (Low/Medium boundary) and **60%** (Medium/High boundary), adjustable through the dashboard interface.

| Probability Range | Risk Category | HR Action Priority |
|---|---|---|
| p < 30% | ■ **Low Risk** | Standard workforce management |
| 30% ≤ p ≤ 60% | ■ **Medium Risk** | Proactive engagement & monitoring |
| p > 60% | ■ **High Risk** | Immediate retention intervention |

# 4. Results

## 4.1 Model Evaluation

All three models were evaluated on the held-out test set (294 records) using five classification metrics. The best model is selected by ROC-AUC — the metric most robust to class imbalance.

| Model | Accuracy | Precision | Recall | F1-Score | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression ■ | 73.8% | 35.0% | **74.5%** | 47.6% | **81.0%** |
| XGBoost | 87.1% | 69.6% | 34.0% | 45.7% | 80.7% |
| Random Forest | 87.1% | 65.5% | 40.4% | 45.7% | 78.9% |

*Caption: ■ = Selected model. Highlighted values — ROC-AUC (orange) and Recall (green) — are the primary selection criteria.*

**Model Selection Rationale:** Logistic Regression was selected as the final model based on its superior ROC-AUC (81.0%) and highest Recall (74.5%). In attrition prediction, recall is the most critical operational metric — missing an at-risk employee (false negative) is typically more costly than an unnecessary retention conversation (false positive). The ROC-AUC of 81.0% indicates strong discrimination between attrition and retention cases across all probability thresholds.

## 4.2 Risk Distribution

Applying the selected model to all 1,470 employees with default thresholds (30% / 60%):

| Risk Category | Count | Percentage | Interpretation |
|---|---|---|---|
| ■ High Risk | 351 | 23.9% | Priority retention intervention |
| ■ Medium Risk | 377 | 25.6% | Monitoring and proactive engagement |
| ■ Low Risk | 742 | 50.5% | Standard workforce management |
| **Total** | **1,470** | **100%** | **Complete Palo Alto Networks dataset** |

The model identifies **351 employees (23.9%)** as High Risk — a focused, actionable cohort for targeted HR retention interventions.

## 4.3 Top Attrition Predictors (Feature Importance)

Feature importance from the trained Logistic Regression model (absolute coefficient magnitude). The top feature's score is normalized to 1.0.

| Rank | Feature | Importance Score | Interpretation |
|---|---|---|---|
| 1 | **MonthlyIncome** | **0.7462** | Lower compensation → higher risk |
| 2 | YearsWithCurrManager | 0.5417 | Manager relationship stability |
| 3 | NumCompaniesWorked | 0.5250 | Prior job-hopping behavior |
| 4 | YearsAtCompany | 0.5008 | Early/late tenure risk |
| 5 | Department | 0.4756 | Role-specific environment |
| 6 | EngagementScore | 0.4044 | Composite satisfaction signal |

| | | | |
|---|---|---|---|
| 7 | OverTime / Workload | 0.3902 | Workload pressure indicator |
| 8 | MaritalStatus | 0.3868 | Life circumstance factor |
| 9 | YearsInCurrentRole | 0.3806 | Role stagnation signal |
| 10 | IncomeExperienceRatio | 0.3640 | Underpaid relative to tenure |

## 4.4 Key Findings

**Finding 1 — Compensation is the Dominant Predictor**
MonthlyIncome is the single most influential predictor (importance: 0.7462). Employees with lower monthly income relative to their cohort exhibit substantially elevated predicted attrition risk. Compensation competitiveness is identified as the primary HR lever.

**Finding 2 — Manager Relationship Quality Matters**
YearsWithCurrManager ranks second (0.5417). Both very short tenure (recent disruption) and extended tenure without advancement contribute to elevated risk patterns.

**Finding 3 — Job Mobility History Predicts Future Mobility**
NumCompaniesWorked ranks third (0.5250). Employees with a history of changing employers are statistically more likely to do so again — reinforcing behavioral alignment in hiring.

**Finding 4 — Tenure and Stagnation are Compounding Risk Factors**
YearsAtCompany (0.5008) and YearsInCurrentRole (0.3806) suggest both early attrition risk (adjustment phase) and late-tenure risk (role stagnation without advancement).

**Finding 5 — Overtime and Low Engagement Compound Risk**
WorkloadStressFlag (0.3902) and EngagementScore (0.4044) both appear in the top 10. Employees simultaneously experiencing high workload and low satisfaction represent a compounding, priority risk profile.

# 5. Discussion

## 5.1 Practical Implications for HR

The system's output enables HR teams to move from broad workforce statistics to targeted, evidence-based retention strategies:

- **High-Risk employees (n=351):** Priority candidates for immediate retention conversations, compensation reviews, and engagement check-ins — particularly those flagged for low income, low engagement, and overtime patterns.
- **Department-level patterns:** The heatmap visualization identifies which departments and job roles have disproportionate concentrations of high-risk employees, enabling targeted resource allocation.
- **What-If Scenario Analysis:** HR partners can model the expected change in a specific employee's attrition probability for hypothetical interventions — providing a quantified basis for retention ROI estimates.

## 5.2 Limitations

- Feature importance does not establish causation; unmeasured confounders (team dynamics, external offers) may also influence attrition.
- The model is trained on a historical snapshot dataset. Workforce conditions evolve and require periodic model re-validation and retraining.
- Satisfaction and engagement variables are self-reported and subject to social desirability bias.
- Default risk thresholds (30% / 60%) should be calibrated to organizational intervention capacity and risk tolerance.

## 5.3 Responsible Use

> **Important Notice**
> This system is designed as a decision-support tool, not an automated personnel decision system. Model predictions must be reviewed by qualified HR professionals, must not be the sole basis for employment-related decisions, and should be audited for differential performance across protected demographic groups.

# 6. Conclusion

This paper presented a machine learning-based employee attrition prediction and risk scoring system for Palo Alto Networks. Through a structured pipeline encompassing data preprocessing, feature engineering, multi-model training, and SHAP-based explainability, the system produces actionable, employee-level attrition risk estimates with a best-model ROC-AUC of **81.0%**.

The primary findings — that compensation, manager relationship quality, prior job mobility, and engagement-workload balance are the strongest predictors of attrition — provide specific, actionable directions for HR retention strategy. The interactive Streamlit dashboard translates these insights into an accessible interface enabling risk exploration, department-level analysis, individual employee

profiling, and what-if scenario simulation — all grounded in actual model predictions rather than fabricated values.

By moving HR analytics from retrospective reporting to **predictive decision intelligence**, this system provides Palo Alto Networks with the capability to identify retention risks earlier, prioritize interventions more effectively, and measure the potential impact of workforce investments — while maintaining appropriate human oversight over final employment decisions.

## 7. Recommendations

**R1 — Compensation Benchmarking**
Conduct an immediate salary equity review for employees in the bottom 25th income percentile within each job level, prioritizing those with high engagement scores — indicating they are committed but potentially underpaid.

**R2 — Manager Continuity Programs**
Identify employees who recently experienced a manager change and schedule structured check-in conversations. Prioritize departments with above-average risk indices.

**R3 — Career Mobility Initiative**
For employees with YearsSinceLastPromotion ≥ 4 and strong performance ratings, initiate structured career development conversations and transparent promotion pathways.

**R4 — Overtime Management**
For High Risk employees flagged for sustained overtime, evaluate workload sustainability and implement targeted workload redistribution before attrition materializes.

**R5 — Quarterly Model Review**
Retrain the pipeline quarterly as new HR data is collected, and monitor model recall on fresh test data to detect performance drift over time.

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